



Mark Dosen <mark@dosen1.com>

RE: Bi-Polar and school air quality

1 message

Ryan Schmit <RSchmit@thendesign.com>

Thu, Nov 12, 2020 at 8:17 AM

To: Mark Dosen <mark@dosen1.com>, Chris Coad <coadc@bbhcsd.org>

Cc: Joelle Magyar <magyarj@bbhcsd.org>, Ellen Kramer <kramere@bbhcsd.org>, Craig Yaniglos <yaniglosc@bbhcsd.org>, Kathleen Mack <mackk@bbhcsd.org>, Brittany Wigman <wigmanb@bbhcsd.org>, Fred Pedersen <pedersenf@bbhcsd.org>, Jeff Crocker <JCrocker@thendesign.com>

Mark,

In speaking with Nate Kobb (Mechanical Engineer), it is our understanding that there shouldn't be a substantial cost increase to install the units later.

The opinion of cost is approximately \$100,000. Of that, \$6,500 is additional design fees and approximately \$15,000 is electrical runs. So, the remaining would run approximately \$78,500. There may be some cost related to contractor remobilization, depending when this work would be performed. That value is difficult to gauge – I will discuss with Eric Gradert to see if we can get an idea for that value.

Thank you,

Ryan Schmit ra, leed ap, ncarb

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From: Mark Dosen <mark@dosen1.com>**Sent:** Wednesday, November 11, 2020 10:34 AM**To:** Ryan Schmit <RSchmit@thendesign.com>; Chris Coad <coadc@bbhcsd.org>**Cc:** Joelle Magyar <magyarj@bbhcsd.org>; Ellen Kramer <kramere@bbhcsd.org>; Craig Yaniglos <yaniglosc@bbhcsd.org>; Kathleen Mack <mackk@bbhcsd.org>; Brittany Wigman <wigmanb@bbhcsd.org>; Fred Pedersen <pedersenf@bbhcsd.org>; Mark Dosen <mark@dosen1.com>**Subject:** Re: Bi-Polar and school air quality

Ryan / Chris - Would there be value in installing the 10 electrical runs (~\$15,000) but holding off on installing the bi-polar units? With electrical in place, how much extra cost would there be to add the bi-polar units later after the building opens?

Thanks...Mark

On Tue, Nov 10, 2020 at 9:21 AM Joelle Magyar <magyarj@bbhcsd.org> wrote:

Here are responses:

- Will selecting Bi-Polar allow us to use a smaller, less costly HVAC system? If so, what are the one-time savings?
 - Bi-polar ionization, if included in the design from the beginning, allows you to reduce the total amount of fresh air being brought into the building which in turn reduces the load. It is likely too late at this juncture to downsize the major HVAC equipment as it's been ordered long ago and would require a major design change.
- If this requires a separate electrical run, what is the cost for the run?

- I would estimate \$1500/unit. There are about 10 units.
- Do we believe Bi-Polar will net us ongoing operational savings and, if so, how much?
 - It can net overall operational savings if we re-calculate the fresh air requirements and redesign the air balance to reduce the total ventilation load. The systems currently have energy recovery wheels and demand control ventilation in place which already offset a typical ventilation load to the building but a conservative estimate could be \$5-10K per year of energy savings with additional reduced energy loads.

On Tue, Nov 10, 2020 at 7:58 AM Ellen Kramer <kramere@bbhcsd.org> wrote:

I second Mark's request. Especially since I just drove by our sad-looking tennis courts again this morning.

On Mon, Nov 9, 2020 at 11:38 AM Mark Dosen <mark@dosen1.com> wrote:

I looked through the material and, unless I missed it, I saw no actual results of lower illness or absenteeism from using this technology. It still all appears theoretical.

There are claims that this could reduce upfront costs by allowing for a smaller, less costly HVAC system and that it could reduce operational costs. Per Ryan, it also appears separate electrical may need to be run. Can we get feedback on these questions:

- Will selecting Bi-Polar allow us to use a smaller, less costly HVAC system? If so, what are the one-time savings?
- If this requires a separate electrical run, what is the cost for the run?
- Do we believe Bi-Polar will net us ongoing operational savings and, if so, how much?

We are all familiar with our long list of PI needs and our difficulty of identifying funding to address them. Every dollar of savings we can glean from the new school can be used to address our PI backlog in line with what we told our voters when we placed the bond issue on the ballot. It is a zero-sum game, money spent on Bi-Polar cannot be spent on roofs. Let's get some more cost numbers and discuss this on 11/18.

Thanks...Mark

On Fri, Nov 6, 2020 at 2:30 PM Fred Pedersen <pedersenf@bbhcsd.org> wrote:

Excellent information Craig!

Thanks,

Fred

On Fri, Nov 6, 2020, 9:54 AM Craig Yaniglos <yaniglosc@bbhcsd.org> wrote:

This as well is some great information!

Attached.

On Fri, Nov 6, 2020 at 8:43 AM Joelle Magyar <magyarj@bbhcsd.org> wrote:

Here is the information you requested. I just want to remind the Board this is not an EXTRA cost from the standpoint that this will be covered under Bid- day savings and/or contingency already built into the project. If the \$100,000 is not spent on this, it will be spent on something else either in the project or to another project in the district. We are not spending additional tax payer dollars. This is simply authorizing a change order that is over the \$50,000 threshold that Craig and I can approve.

Thanks,

Joelle

----- Forwarded message -----

From: **Ryan Schmit** <RSchmit@thendesign.com>

Date: Fri, Nov 6, 2020 at 8:38 AM

Subject: Bi-Polar and school air quality

To: Craig Yaniglos <yaniglosc@bbhcsd.org>

Cc: Christopher Coad <coadc@bbhcsd.org>, Joelle Magyar <magyarj@bbhcsd.org>, Jeff Crocker <JCrocker@thendesign.com>

Good Morning Craig,

Listed below are several references.

We have looked into the electrical requirements. It appears the units do pull around 10 Amps for the larger ones. That said, it is looking unlikely that we will be able to utilize the service outlets. Thorson Baker is still confirming this. Typically the units do get wired separately.

Please review and let us know if you have any questions.

Thank you,

Ryan Schmit ra, leed ap, ncarb



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Here are some references for reduced absenteeism and better indoor air quality for schools as well as several school links showing the use of the technology for better IAQ and to combat Covid.

http://www.bioclimatic.com/tech_ionization.htm

<https://blog.plasma-air.com/the-relationship-between-classroom-air-quality-and-student-performance/>

<https://www.goldenrams.com/site/default.aspx?PageType=3&DomainID=1&ModuleInstanceID=1030&ViewID=6446EE88-D30C-497E-9316-3F8874B3E108&RenderLoc=0&FlexDataID=16551&PageID=1>

<https://www.cpalions.org/about/news-events/news-page/~board/all-school-news/post/cpa-installs-needlepoint-bipolar-ionization-technology>

BPI is a cost-effective solution to deliver a healthier, more pleasing indoor environment. Furthermore, designers of the HVAC system may be able to reduce the amount of the outside air ventilation load while maintaining or actually improving indoor air quality, leading to significant heating and cooling energy cost savings.

Benefits:

- Mitigates odors
- Reduces many gaseous contaminants
- Controls VOCs
- Deactivates bacteria, viruses, mold, and fungi
- Reduces ventilation requirements, operating, and maintenance costs
- Easily integrated in to HVAC systems



By improving the school environment BPI can help improve attendance, learning, and test scores.

Nate Kobb, PE, CxA, LEED AP

Principal | Mechanical Engineering + Commissioning



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Joelle Magyar, Superintendent

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--

Craig Yaniglos, BSE

Chief Financial Officer

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